

Mathematics Advanced

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| General Instructions | <ul style="list-style-type: none">• Reading time – 10 minutes• Working time – 3 hours• Write using black pen• Calculators approved by NESA may be used• For questions in Section II, show relevant mathematical reasoning and/ or calculations |
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| Total Marks:
100 | <p>Section I – 10 marks (pages 1–3)</p> <ul style="list-style-type: none">• Attempt Questions 1–10• Allow about 15 minutes for this section <p>Section II – 60 marks (pages 4–25)</p> <ul style="list-style-type: none">• Attempt Questions 11–29• Allow about 2 hours and 45 minutes for this section |
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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

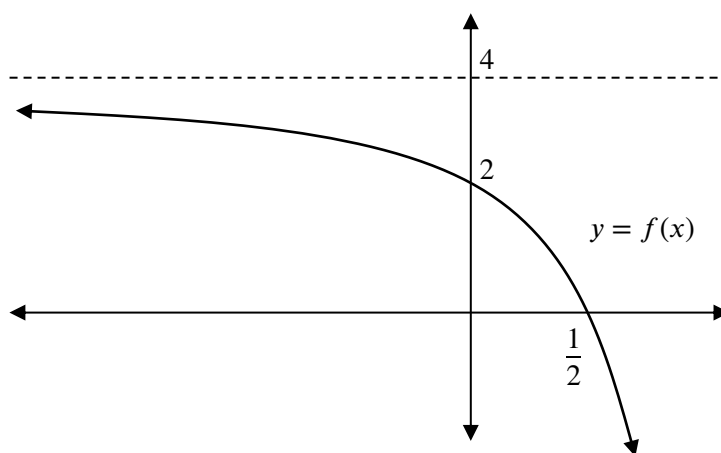
- 1 What is the y -intercept on the graph of $y = -2(3 - x)(x + 5)$?
 - A. 3
 - B. -5
 - C. -10
 - D. -30

- 2 The random variable X is distributed such that $X \sim N(8.3, 0.64)$. In what range do the middle 68% of scores lie?
 - A. $[7.99, 8.64]$
 - B. $[7.66, 8.94]$
 - C. $[7.02, 9.58]$
 - D. $[6.38, 10.22]$

- 3 A fair die is rolled twice, and the numbers obtained on each roll are multiplied. What is the probability that this product is an odd number?
 - A. $\frac{1}{4}$
 - B. $\frac{3}{8}$
 - C. $\frac{1}{2}$
 - D. $\frac{3}{4}$

- 4 An arithmetic sequence with a common difference of 8 between successive terms has a 7th term which is 5 times its first term. What is the 10th term in the sequence?
 - A. 40
 - B. 80
 - C. 84
 - D. 87

- 5 What is the range of $y = \ln[20 + 2e^{-3x}]$?
- $y < \ln 18$
 - $y > \ln 20$
 - $y > \ln 22$
 - $y \geq \ln 22$
- 6 For what values of x is the graph of $y = e^{2x} + 3x - 5x^2$ concave up?
- $x > \ln \frac{5}{2}$
 - $x < \ln \frac{2}{5}$
 - $x > \frac{1}{2} \ln \frac{5}{2}$
 - $x < \frac{1}{2} \ln 5$
- 7 The diagram below shows the graph of $y = f(x)$, where $f(x)$ is an exponential function.
Which of the following is the equation of $f(x)$?



- $f(x) = 4 - 2e^x$
- $f(x) = 4 - 2^{2x+1}$
- $f(x) = 4 - 2^{x+1}$
- $f(x) = 4 + 2^{x+1}$

- 8 A particle moves with displacement given by $x = \frac{5}{1+2t} + kt$, where k is an unknown constant.

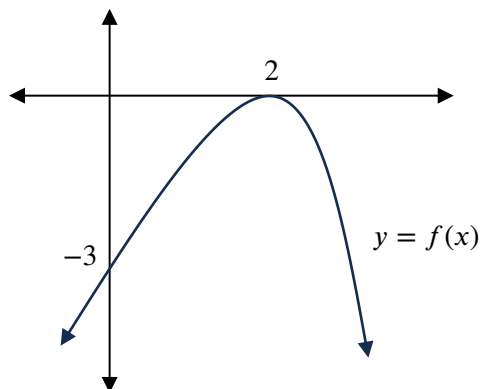
Given that it is initially stationary, which of the following is its displacement at $t = 1$?

- A. 5
- B. 10
- C. $\frac{20}{3}$
- D. $\frac{35}{3}$

- 9 Find $\int_0^{\frac{\pi}{3}} \frac{1}{1-\sin^2 x} dx$

- A. $\frac{1}{\sqrt{2}}$
- B. $\frac{1}{\sqrt{3}}$
- C. 1
- D. $\sqrt{3}$

- 10 A graph of the function $f(x)$ is shown below. It has a single stationary point at $(2,0)$ and is defined for all real x :



Define the function $g(x) = f(f(x))$. Which of the following is false?

- A. $g(2) = -3$
- B. $g(x)$ is negative for the entirety of its domain
- C. $g'(2) = 0$
- D. $g'(x)$ is negative for $x < 2$

Section II

90 marks

Attempt Questions 11–29

Allow about 2 hours and 45 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (4 marks)

The series below is the beginning of an arithmetic sequence of numbers.

7, 11, 15, 19 ...

- a) Find the value of the 45th term in the series.

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- b) Find the sum of the first 28 terms.

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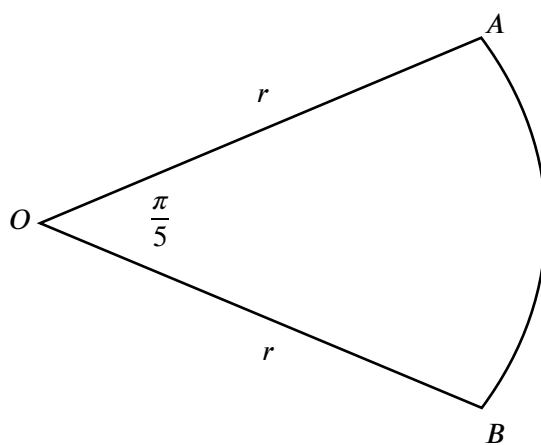
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Question 12 (4 marks)

The circular sector below has an angle of $\frac{\pi}{5}$ at point O , and has a total perimeter of $20 + 2\pi$.



- a) Find the radius r of the sector.

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- b) Find the area of the sector.

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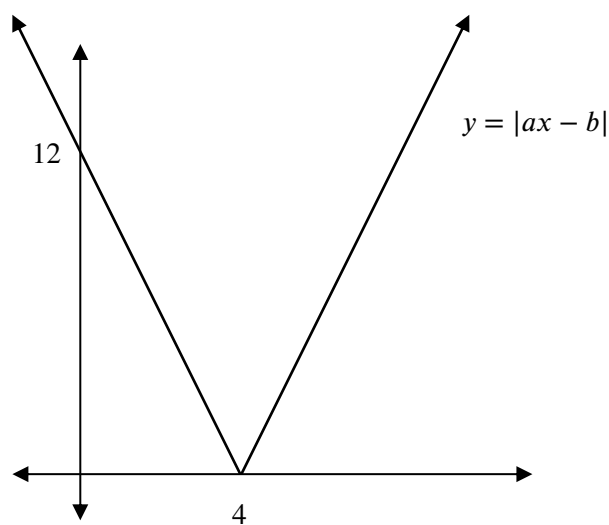
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Question 13 (2 marks)

The diagram below shows the graph of the function $f(x) = |ax - b|$, where a and b are both positive.



Find the values of a and b .

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Question 14 (3 marks)

Solve $2 \cos 3x = \sqrt{3}$ for $0 \leq x \leq \pi$

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Question 15 (4 marks)

80 people were asked to choose their preference between vanilla and chocolate ice cream. The two-way table below shows the results.

	Male	Female	Totals
Chocolate		21	
Vanilla	29	13	42
Totals	46		80

a) Fill in the blank spaces in the table. 2

b) Based on the above data, what is the probability that someone is male, given that they like chocolate ice cream? 2

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Question 16 (2 marks)

Differentiate $y = \frac{x^2}{1-e^x}$

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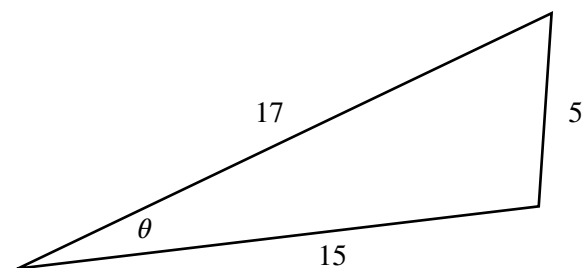
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Question 17 (4 marks)

A triangle is drawn below with sides labelled as shown:



- a) Find the angle θ 2

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- b) Hence find the area of the triangle, correct to 2 decimal places 2

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Question 18 (3 marks)

The function f is defined piecewise as follows:

$$f(x) = \begin{cases} 1 - \frac{3}{4}x^2, & 0 < x < 1 \\ m(x - 3), & 1 \leq x < 3 \\ 0, & \text{elsewhere} \end{cases}$$

Find the value of m such that $f(x)$ is a probability density function.

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Question 19 (4 marks)

The typing speeds of a large group of computer users was observed, and it was concluded that typing speed can be modelled with a normal distributed random variable X , which has a mean of 46 WPM (words per minute)

- a) Given that 16% of scores are above 60, find the standard deviation of X

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- b) Find the probability that a person's typing speed is above 70 WPM. You may use the information in the table below:

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$P(Z \leq z)$ for the standard normal distribution $N(0, 1)$

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706

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Question 20 (8 marks)

Consider the function $f(x) = \frac{x}{x^2-4}$

- a) State the equations of all asymptotes on the graph of $y = f(x)$

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Question 20 continues on the next page

- b) Show that $f'(x) = -\frac{x^2+4}{(x^2-4)^2}$, and hence show that there are no stationary points on the graph of $y = f(x)$

[illegible]

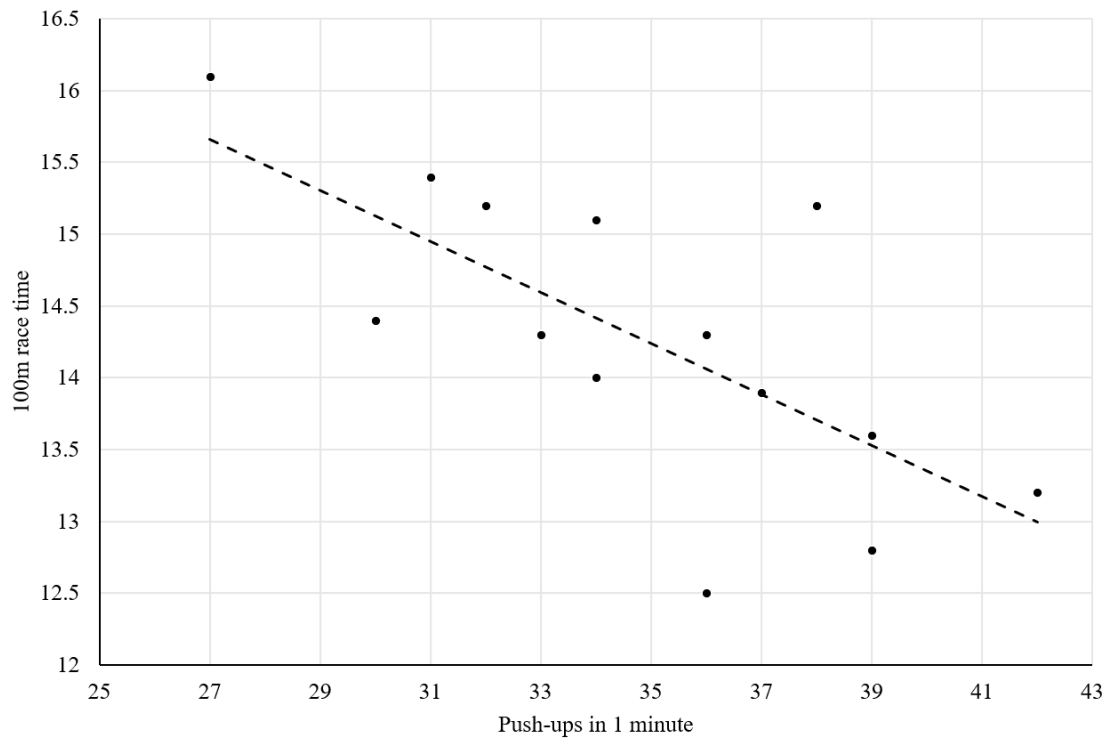
- c) By considering a transformation, or otherwise, sketch the graph of $y = \frac{x+1}{(x-1)(x+3)}$

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Question 21 (3 marks)

The coach of a local Under-19's soccer team records training statistics for the 15 boys in the team. He counts the number of push-ups (a type of arm exercise) each player can do in one minute, and times how long each takes to run 100m.

The graph and trendline show the results - the equation of the trendline is $y = -0.18x + 20.47$, with a R-coefficient of -0.70 .



Comment on the statistical results, and suggest why the results are distributed as they are.

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Question 22 (6 marks)

Consider the integral $\int_0^{\frac{3}{2}} \frac{x}{1+x^2} dx$

- a) Use the trapezoidal rule with 5 function values to estimate the integral

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- b) Use calculus to find the exact value of the integral

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- c) Is the graph of $y = \frac{x}{1+x^2}$ concave up or down for the domain $0 < x < \frac{3}{2}$? Provide a brief explanation for your answer.

Note: You may assume that the concavity does not change in this domain.

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Question 23 (6 marks)

A frozen chicken is left on a kitchen benchtop to thaw. The temperature of the chicken is modelled by the equation:

$$T(t) = 23 - Be^{-kt}$$

Where T is temperature, t is time in minutes, and B and k are constants.

The temperature of the chicken when it is taken out of the freezer is -10°C .

- a) Find the value of B **1**

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- b) The chicken reaches a temperature of 0°C after 3 minutes of staying outside. Find the value of k . **2**

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Question 23 continues on the next page

- c) Find the time when the temperature is increasing at an instantaneous rate of $2^{\circ}\text{C}/\text{min}$, correct to the nearest second.

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Question 24 (5 marks)

The temperature in a classroom is modelled by the equation $T = A \sin\left(\frac{\pi}{12}t\right) + C$, where A and C are constants, and t is number of hours past 6:00am on Monday.

The temperature in the classroom varies between 19° and 31°

- a) Find the values of A and C **2**

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- b) The class has a pet lizard named Keith, who lives in a glass enclosure on a bench at the side of the room. The enclosure contains a small resting place, from which Keith only emerges if the temperature is above 28° .

Find the times during which Keith comes out of his rest area on Monday. **3**

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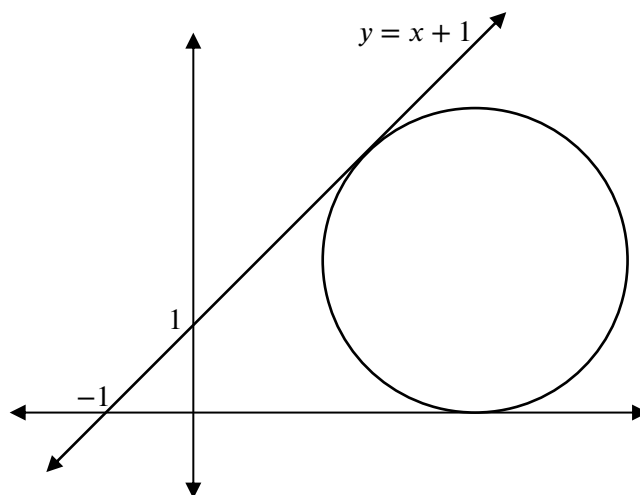
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Question 25 (6 marks)

The diagram shows a circle which is tangent to both the x -axis and the line $y = x + 1$. The circle has a radius of 3 units, and is defined by the equation $(x - k)^2 + (y - h)^2 = c$



- a) State the values of h and c

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Question 25 continues on the next page

b) Find the equation of the circle.

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Question 26 (6 marks)

Consider the equation $\log_2 x + \log_x 8 = 4$

- a) Show that $(\log_2 x)^2 + 3 = 4 \log_2 x$ **3**

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- b) Hence find all possible values for x **3**

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Question 27 (5 marks)

In a reducing-balance loan with monthly compounding, an amount $\$P$ is borrowed at a rate of $R\%$ per annum, and repayments of $\$M$ are made every month, starting one month after the loan is taken out. After n repayments, the outstanding debt is given by:

$$P \left(1 + \frac{0.01R}{12}\right)^n - M \left(1 + \left(1 + \frac{0.01R}{12}\right) + \left(1 + \frac{0.01R}{12}\right)^2 + \dots + \left(1 + \frac{0.01R}{12}\right)^{n-1}\right)$$

Daniel plans to take out a loan for a business venture. After reviewing his financial situation, he concludes that he can reserve at most \$250 in monthly loan repayments.

The bank from which he decides to obtain the loan is offering a compounding rate of 3.6% per annum, compounding monthly. He also decides that he wants to spend a maximum of 3 years paying off the loan, i.e. he wants to complete the payment of the loan in 36 instalments

- a) Find the maximum amount he can borrow.

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Question 27 continues on the next page

- b) After 21 repayments have been made, the bank decides to up the rate to 6.3% per annum. Daniel wants to stick to the limit he set earlier, and tries to repay the loan before the original 3 years elapses. Fortunately enough, his business venture succeeded, allowing him to make larger repayments.

His outstanding debt when the interest rate changes is \$3662 (do NOT prove this)

Find the new monthly payment he should be making so that his loan is paid off by the 3 years, to the nearest cent. **2**

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Question 28 (4 marks)

A particle is initially at the origin, moving in the positive direction. It returns to the origin after 8 seconds. Throughout its motion, the particle experiences a constant acceleration of -4 m s^{-2} .

Find the furthest distance the particle attains from the origin during these 8 seconds.

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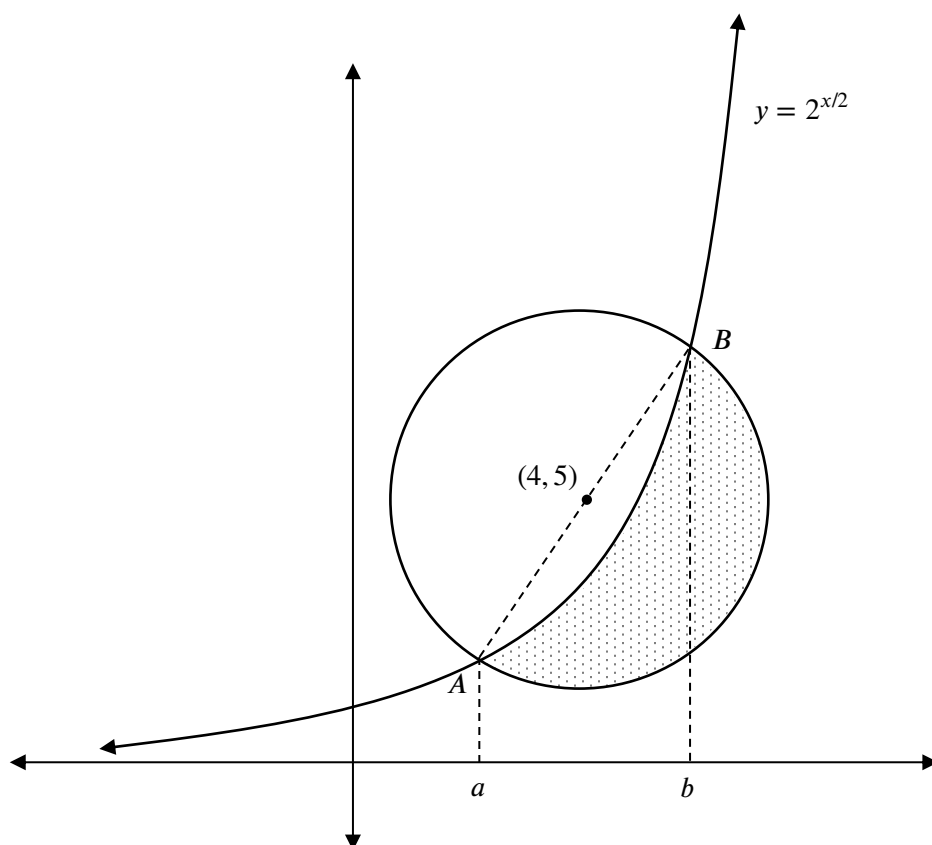
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Question 29 (11 marks)

A circle centred at $(4, 5)$ intersects with the graph $y = 2^{x/2}$ at points A and B , with x -values a and b respectively.

It is given that AB is a diameter of the circle.



- a) Show that $a + b = 8$ and $2^{a/2} + 2^{b/2} = 10$

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Question 29 continues on the next page

b) Hence show that

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$$(2^{a/2})^2 - 10 \times 2^{a/2} + 16 = 0$$

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c) By letting $u = 2^{a/2}$ in the above equation, or otherwise, find a and b , and hence find the equation of the line AB (3)

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Question 29 continues on the next page

d) Find the shaded area in exact form.

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